



**FACULTY OF ENGINEERING, BUILT ENVIRONMENT, AND
INFORMATION TECHNOLOGY (FOEBEIT)**

BACHELOR OF COMPUTER SCIENCE (ODL)

BCL1233 – SYSTEM ANALYSIS AND DESIGN

ASSIGNMENT 3

Logic and Data Modeling for a Hybrid Work Monitoring System

INSTRUCTIONS TO CANDIDATE:

1. This is an **INDIVIDUAL** assessment.
2. This assignment contributes **20%** of the total course mark.
3. Submit your completed assessment via **SPARK by 11:59 PM on 25 July 2026.**

WARNING

You are required to submit your work within the bounds of the University Infringement of Assessment Regulations. Plagiarism, paraphrasing and downloading large amounts of information from external sources will not be tolerated and will be dealt with severely.

SEGi University, Kota Damansara takes allegations of plagiarism very seriously. Submissions involving plagiarism will be marked but given zero mark. Plagiarism is the attempt to pass off the work of another as your own. Information taken from the work of others should be acknowledged by reference to obviate the charge of copying.

The work must be entirely your own. You will receive ZERO (0) for work that is not your own. The safety of your assessments is your responsibility. You must not permit another student access to your work.

OBJECTIVE

This assessment aims to evaluate students' ability to:

1. Identify and analyse problems in an existing system
2. Determine system objectives based on user requirements
3. Identify stakeholders and their roles
4. Specify functional and non-functional system requirements
5. Apply appropriate requirement gathering techniques
6. Demonstrate understanding of requirement analysis in system development

CASE STUDY: Hybrid Work Monitoring System

A technology company has adopted a hybrid working model where employees work either from home or the office. Currently, the company uses manual tools such as spreadsheets, chat messages, and emails to track employee attendance, task updates, and work progress. This approach has led to several issues, including:

1. Inaccurate attendance tracking
2. Lack of real-time work visibility
3. Difficulty in monitoring task progress
4. Communication delays between employees and managers
5. No centralized reporting system

To address these issues, the company plans to develop a **Hybrid Work Monitoring System**. The proposed system will:

1. Allow employees to digitally check in and check out
2. Record work location (remote or office)
3. Enable task assignment and tracking
4. Provide real-time work status updates
5. Generate attendance and productivity reports
6. Send notifications for tasks and meetings
7. Provide a management dashboard

TASK DESCRIPTIONS

The organisation requires a Hybrid Work Monitoring System to manage employee work schedules, attendance, task progress, and performance monitoring. For this assignment, students are required to focus on the following business process:

Hybrid Work Schedule Approval Process
The process consists of the following activities: 1. Employee submits a hybrid work request. 2. The system validates the request. 3. The supervisor reviews the request. 4. The supervisor approves or rejects the request. 5. The system updates the employee's work schedule. 6. The system sends a notification to the employee. The following business rules apply: 1. Employees may work remotely for a maximum of three days per week. 2. Requests must be submitted at least two days before the requested date. 3. Supervisor approval is required for all requests. 4. Each department must maintain at least 50% on-site staff availability.

REPORT REQUIREMENTS

Students must prepare a structured report and respond to ALL sections below.

1. Logic Modeling (40 marks)

- a) Analyse the business rules for the Hybrid Work Schedule Approval Process.
- b) Develop **ONE (1)** of the following logic models:
 - Decision Table; or
 - Decision Tree; or
 - Structured English.
- c) Clearly identify:
 - Conditions
 - Actions
 - Decision outcomes
- d) Provide a brief explanation of the selected logic model.

2. Entity Relationship Diagram (ERD) (35 marks)

- a) Identify the entities, attributes, and relationships related to the Hybrid Work Schedule Approval Process.
- b) Develop an ERD that includes:
 - Entities
 - Attributes
 - Primary keys
 - Foreign keys
 - Relationship cardinalities
- c) Ensure consistency between the ERD and the process models developed in Assignment 2.

3. Data Dictionary (15 marks)

Develop a data dictionary for the entities identified in the ERD. The data dictionary should include:

- Attribute name
- Description
- Data type
- Key type (Primary Key/Foreign Key)
- Validation rules

4. Model Validation and Integration (10 marks)

Write a brief discussion (200–300 words) explaining:

- a) Discuss the role of the logic model in supporting the Hybrid Work Schedule Approval Process.
- b) Describe the relationship between the ERD, data dictionary, and the requirements identified in Assignment 1.
- c) Explain the relationship between the models developed in this assignment and the process models created in Assignment 2.

EVALUATION CRITERIA

Assessment Component	Description	Mark Allocation
Logic Modeling	Accuracy and completeness of conditions, actions, and decision outcomes.	40%
Entity Relationship Diagram (ERD)	Correct identification of entities, attributes, keys, relationships, and cardinalities.	35%
Data Dictionary	Completeness and consistency of data definitions.	15%
Model Validation and Integration	Ability to relate models to previous assignments and demonstrate consistency.	10%
TOTAL:		100%

SUBMISSION REQUIREMENTS

Students are required to submit the following upon completion of this assessment:

1. A complete report in **Word or PDF format**.
2. Structured answers covering all tasks and questions.
3. Clear headings and proper formatting for each section.
4. Optional diagrams or illustrations (if needed for explanation).
5. Submission must be uploaded via **SPARK** on or before the given deadline.